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TOWARDS TRUSTWORTHY AI IN HEALTHCARE

CALIBRATION, EVIDENCE-BASED EVALUATION, AND THE CLINICAL UTILITY OF THE ADNEX MODEL

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We shall try not to use statistics as a drunken man uses lamp-posts, for support rather than for illumination.

– Andrew Lang, Scottish poet and novelist

Contents

Glossary

Terms shown in bold at first mention are defined in this glossary. Definitions in this glossary are provided in the context of clinical prediction modelling and are tailored to the scope of this thesis.

Apparent performance Model performance assessed in the exact same data (i.e. training data) used for model development.

Bootstrap A resampling method for assessing model performance by repeatedly sampling with replacement from the original dataset to create multiple bootstrap samples, fitting the model on each sample, and evaluating performance on the out-of-bag observations.

Calibration The agreement between predicted probabilities and observed outcome proportion.

Calibration intercept The parameter that quantifies systematic overestimation or underestimation of predicted risks. A positive calibration intercept indicates systematic underestimation of risk, while a negative calibration intercept indicates systematic overestimation of risk. Analogous to the Observed-to-expected ratio.

Calibration plot A graphical method for assessing calibration by plotting observed outcome proportions against predicted probabilities. It is flexible if smoothing is applied often using LOESS, binning is applied, or a logistic calibration curve with splines is fitted.

Calibration slope The parameter that quantifies whether predicted risks are too extreme or too moderate. It is estimated by regressing the observed outcomes on the predicted log-odds of the outcome. A calibration slope less than 1 indicates that predicted risks are too extreme, while a calibration slope greater than 1 indicates that predicted risks are too moderate.

Case mix The distribution of patient characteristics and risk levels within a study population.

Clinical prediction model A statistical model that combines multiple predictors to estimate the probability of a clinical outcome for an individual patient.

Clinical utility The extent to which using model predictions improves clinical decision-making and patient outcomes, typically assessed with decision-analytic measures such as net benefit.

Cross-validation A resampling method for assessing model performance by partitioning the data into complementary subsets, training the model on one subset and validating it on the other subset, and repeating this process multiple times.

Decision curve analysis A method to evaluate clinical usefulness of a prediction model across decision thresholds by plotting net benefit of model-guided decisions versus default strategies of treating all or treating none of the patients.

Decision threshold The risk probability above which a clinical action is recommended.

Diagnosis The process of determining the presence or absence of a disease or condition based on clinical information, which may include symptoms, signs, and model results.

Discrimination The ability of a prediction model to assign higher predicted risks to individuals with the event than to those without the event.

Empirical Bayes estimates Cluster-specific estimates from mixed models that are shrunk toward the overall mean using estimated variance components.

Evidence-based medicine An approach to medical practice intended to optimize decision-making by emphasizing the use of evidence from well designed and conducted research.

External validation Evaluation of a prediction model in data that are independent from the model development data.

False negative A patient with the outcome who is incorrectly identified as negative by a prediction model or decision rule.

False positive A patient without the outcome who is incorrectly identified as positive by a prediction model or decision rule.

Heterogeneity Variation in model performance.

Hyperparameter tuning The data driven process of selecting the optimal values for hyperparameters in a machine learning model or AI algorithm, often using resampling methods such as cross-validation to evaluate performance across different hyperparameter combinations.

Internal validation Assessment of model performance in the development dataset using resampling methods such as bootstrapping or cross-validation.

LOESS Locally Estimated Scatterplot Smoothing, a non-parametric method for fitting a smooth curve to data by fitting simple models to localized subsets of the data.

Logistic calibration A method for assessing calibration by fitting a logistic regression model with the observed outcome as the dependent variable and the predicted log-odds of the outcome as the independent variable. The intercept and slope of this model are the calibration intercept and slope, respectively.

Missing at random A missing-data mechanism where the probability that a value is missing may depend on observed data, but not on the unobserved value itself after conditioning on observed data.

Missing completely at random A missing-data mechanism where the probability that a value is missing does not depend on observed data or unobserved data. Under MCAR, complete cases are a random subsample of all cases.

Missing not at random A missing-data mechanism where the probability that a value is missing depends on the unobserved value itself, even after conditioning on observed data.

Mixed effects model A regression model that combines fixed effects and random effects to account for clustered data or hierarchical data structures.

Net benefit A decision-analytic measure that combines true positives and false positives using a chosen decision threshold. Mathematically defined as: $\frac{\text{TP}}{N} - \frac{\text{FP}}{N} \times \frac{\text{threshold}}{1-\text{threshold}}$, where TP is the number of true positives, FP is the number of false positives, and N is the total number of patients.

Observed-to-expected ratio A calibration measure comparing the total number of observed events to the total number of expected events from the model. A value of 1 indicates perfect agreement in the mean. Values less than 1 indicate overestimation of risk, while values greater than 1 indicate underestimation of risk.

Overfitting A phenomenon where a model captures noise or idiosyncrasies in the development data, leading to poorer performance in new data.

Prediction interval An interval that represents the expected range of true model performance in an hypothetical new cluster.

Predictor A variable used in a prediction model to estimate the probability of an outcome. Also known as a feature, covariate, or independent variable.

Prognosis The process of predicting the future course or outcome of a disease or condition based on clinical information, which may include symptoms, signs, and model results.

Properness A property of a performance measure that is optimized when the true probabilities are used.

Random effects meta-analysis A meta-analytic framework that allows true effects or performance measures to vary across clusters.

Random intercept A random effect that allows the baseline outcome level or risk to vary across clusters while keeping predictor effects fixed.

Random slope A random effect that allows predictor effects to vary across clusters.

Restricted cubic splines A flexible regression approach that models non-linear relationships by joining piecewise cubic polynomials smoothly at predefined knot locations while constraining the function to be linear in the tails.

Shrinkage The reduction of regression coefficients towards 0 to improve predictions in patients outside the development sample. In clustered data, shrinkage can also refer to the reduction of cluster-specific estimates towards the overall mean using empirical Bayes estimates from mixed models.

True negative A patient without the outcome who is correctly identified as negative by a prediction model or decision rule.

True positive A patient with the outcome who is correctly identified as positive by a prediction model or decision rule.

Underfitting A phenomenon where a model is too simple to capture relevant signal in the data, resulting in poor predictive performance.

Variable selection The process of choosing a subset of relevant features for use in model construction.

Violin plot A graphical method for displaying the distribution of a dataset, combining a box plot with a kernel density plot to show the full distribution of the data.

List of Acronyms

2MA-C Two-stage meta-analysis calibration

ADEMP Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures

ADNEX Assessment of Different NEoplasias in the adneXa

AI artificial intelligence

AUC area under the receiver operating characteristic curve

AUROC area under the receiver operating characteristic curve

BD Benign Descriptor

BIC Bayesian information criterion

CA-125 cancer antigen 125

CCA complete case analysis

CG-C Clustered group calibration

CHARMS CHecklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies

CI confidence interval

CIT conditional inference tree

CPM clinical prediction model

CRASH Clinical Randomisation of an Antifibrinolytic in Significant Haemorrhage

CrI credible interval

DCA decision curve analysis

DGM data-generating mechanism

DL DerSimonian-Laird

DU decision uncertainty

ECI estimated calibration index

EIDM evidence-informed decision-making

EPC events per cluster

EPP events per potential predictor

EPV events per variable

EQUATOR Enhancing the QUAlity and Transparency Of health Research

ESGE European Society for Gynaecological Endoscopy

ESGO European Society of Gynaecological Oncology

GCS Glasgow Coma Scale

GLMM generalized linear mixed model

GRADE Grading of Recommendations Assessment, Development and Evaluation

HE4 human epididymis protein 4

HKSJ Hartung-Knapp-Sidik-Jonkman

ICC intra-cluster correlation

IOTA International Ovarian Tumour Analysis

IPD individual patient data

IST International Stroke Trial

ISUOG International Society of Ultrasound in Obstetrics and Gynecology

JAGS Just Another Gibbs Sampler

LR logistic regression

LR1 IOTA logistic regression model 1

LR2 IOTA logistic regression model 2

MAR missing at random

MCAR missing completely at random

MDR Medical Device Regulation

MIX-C Mixed model calibration

MLR multinomial logistic regression

MNAR missing not at random

MSCE mean squared calibration error

MSE mean squared error

NB net benefit

NPV negative predictive value

O-RADS Ovarian-Adnexal Reporting and Data System

O:E observed-to-expected ratio

OSF Open Science Framework

PDI polytomous discrimination index

PET probability estimation tree

PI prediction interval

PM Paule-Mandel

PPV positive predictive value

PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analysis

PROBAST Prediction model Risk Of Bias ASsessment Tool

PROSPERO International prospective register of systematic reviews

QUADAS Quality Assessment of Diagnostic Accuracy Studies

REML restricted maximum likelihood

RF random forest

RMI Risk of Malignancy Index

ROC receiver operating characteristic

ROMA Risk of Ovarian Malignancy Algorithm

RU relative utility

SaMD Software as a Medical Device

SNR signal-to-noise ratio

SRRisk Simple Rules Risk model

STRATOS STRengthening Analytical Thinking for Observational Studies

TBI traumatic brain injury

TRIPOD Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

TT test tradeoff

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Part I

Introduction and research objectives

Chapter 1

Introduction

NB

1.1 Background

Patient healthcare management involves a series of complex decisions that clinicians must make to provide optimal care. Determining which decision to make is often not straightforward, as it requires weighing the potential benefits and harms of different options in an environment with limited time and resources. **Evidence-based medicine** provides a framework for such informed decision-making, integrating the best available research evidence with clinical expertise and patient values [1]. Such evidence-informed decision-making (EIDM) is crucial to ensure that patients receive the most appropriate care while minimizing unnecessary interventions and associated costs [2]. In the current artificial intelligence (AI) and big data era, the volume of available data exceeds human processing capacity, and clinicians are increasingly relying on data-driven tools to support EIDM [3].

These data-driven tools, often taking the form of **clinical prediction models** (CPMs), translate complex patient characteristics into probabilistic risk estimates. By quantifying the probability of a specific disease or future health event, these models allow clinicians and patients to make evidence-informed decisions [4]. However, the methodological quality and performance of these models dictate their safety and effectiveness in the real world and can vary significantly depending on the context.

The role of biostatisticians and methodologists in developing and evaluating such models is therefore crucial. From study design to model implementation and even after deployment, rigorous statistical methodology is required to ensure that the developed tools are accurate, reliable, clinically useful, and fair [5]. This thesis explores the intersection of applied clinical prediction for ovarian cancer and statistical methodology, focusing on how we evaluate and improve these data-driven tools to safely support EIDM.

1.2 Clinical prediction models

Clinical prediction models are statistical tools that combine multiple variables, also known as **predictors**, to estimate the probability of a specific outcome for an individual patient [4]. These models are widely used in medicine to assist clinicians in making informed decisions about **diagnosis**, **prognosis**, and subsequent treatment. This thesis will focus mainly on diagnostic prediction models for a binary outcome, which estimate the probability of a condition being present at the time of evaluation, as opposed to prognostic models, which predict future outcomes [6]. While individual diagnostic tests, such as biomarkers or imaging findings, provide information about a single aspect of the patient's condition, prediction models combine multiple tests and patient characteristics into an integrated probability estimate. Therefore, they can estimate risks for individual patients that can guide clinical management. Successful and widely used clinical prediction models include the Framingham Risk Score (cardiovascular disease) [7], the APACHE II score (critical care) [8], and the Assessment of Different NEoplasias in the adneXa (ADNEX) model (ovarian masses) [9], among others.

A key advantage of clinical prediction models over simpler diagnostic rules is that they produce probabilities. Estimating a 1% probability of a condition being present carries a very different clinical implication than a probability of 15%, yet both would be classified as "low risk" under a **decision threshold** of 20%. This distinction matters because patients and clinicians may weigh the consequences of **false positives** and **false negatives** differently depending on the clinical context. Probabilities preserve this nuance and allow the decision threshold to be set according to the specific clinical situation [10]. This enables shared decision-making between clinicians and patients, where treatment decisions can be tailored to a patient's personal risk and preferences [4, 11].

Two key aspects of clinical prediction models are their development and their evaluation or validation. Model development starts with the design of the study and ends when the final model (e.g. the coefficients of a logistic regression) is obtained. It involves the selection of predictors, the definition of the outcome, the handling of missing data, the fitting of the model among other tasks that are described in detailed by Efthimiou and colleagues [12]. However, models need **external validation** to prove their performance. Ideally, this validation is done by external researchers and in external but comparable settings. This evaluation is crucial to determine whether the model can be reliably applied in clinical practice and to identify any potential limitations or biases. External evaluation is not a one-time event, but rather an ongoing monitoring process that should continue throughout the model's lifecycle [13]. Despite the increasing number of clinical prediction models published each year, only a small fraction are externally validated, and even fewer are adopted in clinical practice [14–16]. Bridging this gap requires rigorous performance evaluation, transparent reporting, and sustained monitoring after deployment [13, 17]. Once a model has been validated in several external settings, this evidence can be synthesised in systematic reviews and meta-analyses to provide a comprehensive assessment of the model's performance across different populations and settings and quantify the variability in performance, also known as performance **heterogeneity** [18].

1.2.1 Model performance

Performance is defined as "how well something or someone does a piece of work or activity" (Cambridge Dictionary). For CPMs performance should be evaluated across several domains to perfectly evaluate how well they perform. Van Calster et al. [19] recommend a minimum core set of measures and visualisations that should be reported when validating a prediction model (Table 1.1). These metrics are selected because their value is optimised when the model predicts correct probabilities (i.e., the metric is "**proper**") and because they focus either on purely statistical performance or decision–analytical performance by properly accounting for misclassification costs. The following subsections introduce the recommended domains and their associated measures.

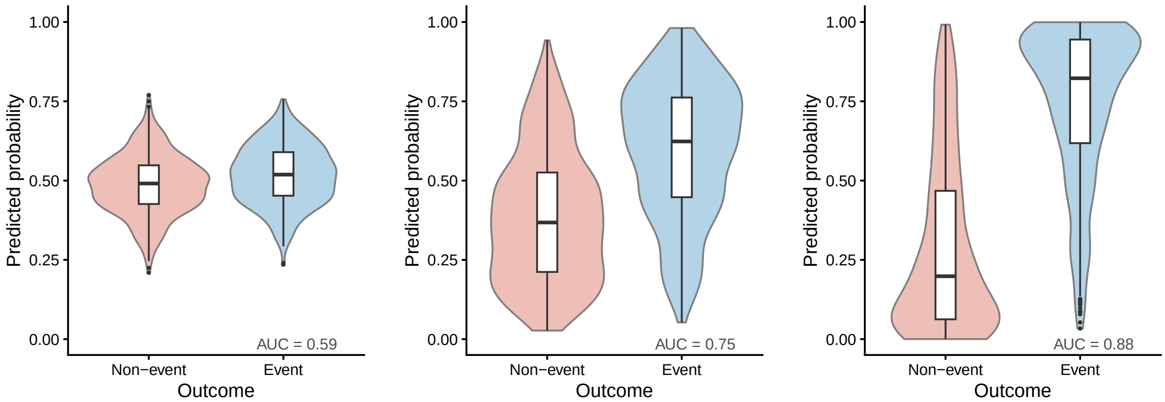
Discrimination

Discrimination refers to a model's ability to distinguish between patients who present the event and those who do not. A model with good discrimination assigns higher predicted probabilities to patients who develop the event than to those who do not. This separation can be visualised using **violin plots** of the predicted probabilities stratified by outcome group (Figure 1.1). The most common summary measure of discrimination is the concordance statistic (c-statistic), equivalent to the area under the receiver operating characteristic curve (AUC) for binary outcomes, which ranges from 0.5 (no discrimination) to 1 (perfect discrimination). AUC and c-statistic can be interpreted

Table 1.1: Recommended core set of performance measures and visualisations for validating CPMs, based on Van Calster et al. [19].

Domain	Measure / visualisation	Ideal value	Recommendation
Discrimination	AUC (c-statistic)	1	Recommended
Calibration	Calibration plot (smoothed)	On diagonal	Recommended
	Calibration intercept (α)	0	Informative
	Calibration slope (β)	1	Informative
	O:E ratio	1	Informative
Clinical utility	Net benefit (decision curve)	Prevalence	Recommended
Overall performance	Risk distribution plots	—	Recommended
Classification	Sensitivity + Specificity	1 each	Descriptive
	PPV + NPV	1 each	Descriptive

as the probability that a randomly selected individual with the event and a randomly selected individual without the event will be correctly ranked by the model, with the former having a higher predicted probability than the latter. For models with multinomial outcomes, the polytomous discrimination index (PDI) is a generalisation of the AUC that quantifies the probability that a randomly selected patient from each outcome group will be correctly ranked by the model [20].



(a) Poor (b) Moderate (c) Good

Figure 1.1: Violin plots of predicted probabilities stratified by outcome group, illustrating three levels of discrimination. Red and blue represent non-events and events, respectively. The AUC increases from poor (a) to good (c) discrimination, reflecting greater separation between the two distributions. Note that poor, moderate, and good discrimination are relative terms that depend on the clinical context and the intended use of the model, rules of thumb about AUC values should be avoided.

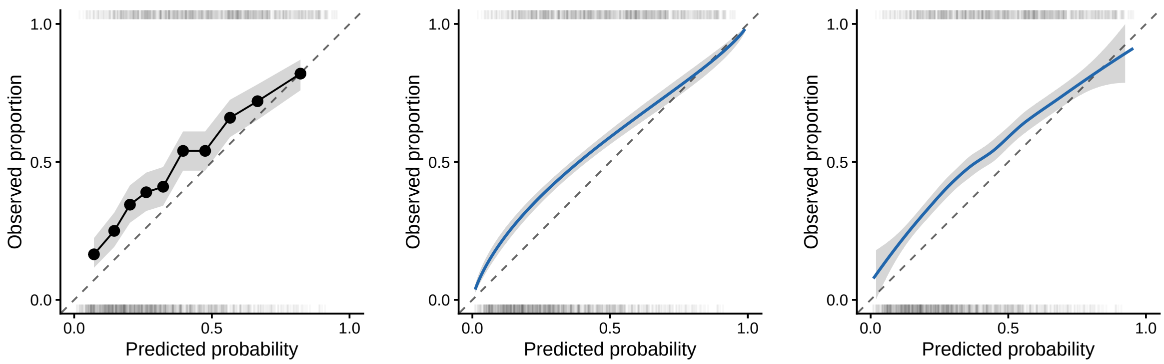
Calibration

Calibration is the agreement between predicted probabilities and observed proportions. In binary outcomes, the observed proportion of events needs to be estimated because the outcome is either 0 (non-event) or 1 (event). This can be done using different methods, such as binned calibration, **logistic calibration**, or **LOESS** smoothing (Figure 1.2). The logistic calibration framework characterises miscalibration through the intercept α and slope β of the model $\text{logit}(p_{\text{obs}}) = \alpha + \beta \cdot \text{logit}(p_{\text{pred}})$, where $\alpha = 0$ and $\beta = 1$ indicate perfect calibration. Different combinations of α and β correspond to distinct types of miscalibration (Figure 1.3).

Van Calster et al. [21] defined a hierarchy of four increasingly strict levels of calibration (Table 1.2). The most basic level is mean calibration, which requires that the average predicted probability equals the observed event rate. The next level is weak calibration, which requires that the calibration intercept α is 0 and the slope β is 1 in the logistic calibration model. Then, moderate calibration requires that the observed event rate equals the predicted probability for every value of the predicted probability, which can be visualised using a flexible **calibration plot**. Flexible calibration plots are created by smoothing the relationship between observed and predicted probabilities using a smoother of choice (e.g. LOESS or **restricted cubic splines**). Finally, strong calibration requires that the observed event rate equals the predicted probability for every covariate pattern, which is an idealised scenario that is practically impossible to achieve.

Table 1.2: Calibration hierarchy for risk models, based on Van Calster et al. [21].

Level	Name	Definition	Evaluation
1	Mean	Average predicted probability equals the observed event rate	O:E ratio
2	Weak	Calibration intercept $\alpha = 0$ and slope $\beta = 1$	Logistic calibration framework
3	Moderate	$P(Y = 1 \mid \hat{p}) = \hat{p}$ for every $\hat{p}$	Flexible calibration plot
4	Strong	$P(Y = 1 \mid \mathbf{X}) = \hat{p}(\mathbf{X})$ for every covariate pattern $\mathbf{X}$	Utopia



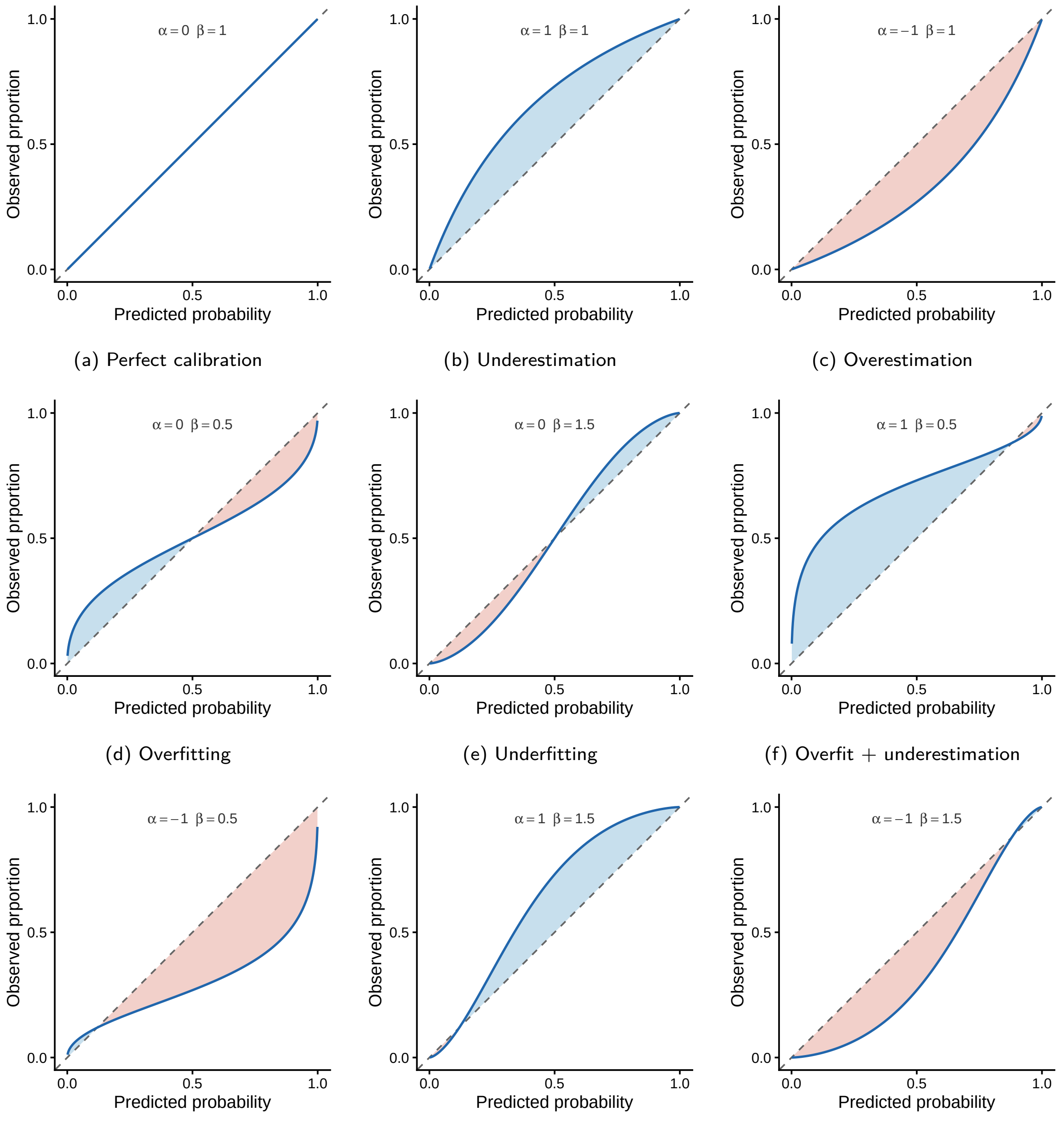
(a) Binned (b) Logistic calibration (c) LOESS

Figure 1.2: Calibration plots for the same simulated data using three methods: (a) binned calibration, (b) logistic calibration, and (c) LOESS smoothed calibration curve. The dashed diagonal represents perfect calibration. Rug plots indicate the distribution of events (top) and non-events (bottom).

Clinical utility

Clinical utility refers to the quality of decisions that are informed by the model's predictions and it is, arguably, the most important aspect of model evaluation. It is a decision-analytic concept therefore it weighs the benefit of the model by the misclassification costs. The main measure of clinical utility is **net benefit** (NB), which quantifies the balance between **true positives** and false positives at a given decision threshold [10]. Net benefit (NB) can be visualised using **decision curve analysis**, which plots NB across a range of decision thresholds to show the clinical value of using the model compared to default strategies of treating all patients or treating no patients or to different models to decide between (Figure 1.4).

The decision threshold is independent of the prediction model and should reflect the clinical context, establishing the relation between the harms of treating the healthy patients and the benefits of treating the diseased patients [22]. This relation can be different across settings, and therefore the decision threshold should be different across settings. For instance, a country with limited access to healthcare might set a higher threshold than a country with abundant resources, because the harm of treating healthy patients is higher in the former. A model that is



(a) Perfect calibration (b) Underestimation (c) Overestimation

(d) Overfitting (e) Underfitting (f) Overfit + underestimation

(g) Overfit + overestimation (h) Underfit + underestimation (i) Underfit + overestimation

Figure 1.3: Types of miscalibration characterised by the calibration intercept α and slope β in the logistic calibration framework $\text{logit}(p_{\text{obs}}) = \alpha + \beta \cdot \text{logit}(p_{\text{pred}})$. Blue shading indicates underestimation (observed $>$ predicted); red shading indicates overestimation (predicted $>$ observed). The dashed diagonal represents perfect calibration ($\alpha = 0$, $\beta = 1$).

clinically useful across a wide range of thresholds is more likely to be adopted in practice, as it can accommodate different clinical contexts and preferences [19].

Clinical utility determines the real-world impact of a prediction model, as it directly relates to patient outcomes and decision-making [23]. A model with excellent discrimination and calibration may still have limited clinical utility if the predicted probabilities do not lead to better decision-making. Conversely, a model with moderate discrimination but good calibration and high NB at relevant thresholds can be highly valuable in practice. Therefore, evaluating clinical utility is essential to ensure that prediction models not only perform well statistically but also translate into meaningful improvements in patient care.

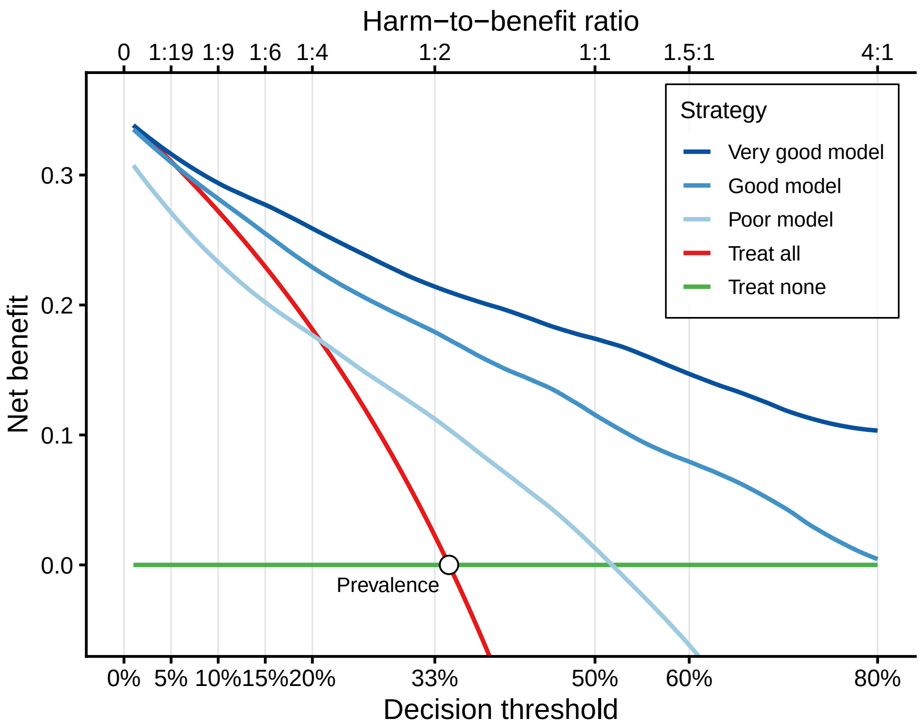


Figure 1.4: Decision curve analysis for three simulated models with varying performance. The very good model (dark blue) and good model (medium blue) provide NB above both default strategies across a wide range of thresholds. The poor model (light blue) is worse than treating all patients at low thresholds and worse than treating none at higher thresholds, therefore it can be harmful for patients. The top axis shows the corresponding harm-to-benefit ratio for each threshold probability.

1.2.2 Developing a prediction model

Developing a clinical prediction model involves a sequence of critical methodological steps, that are well explained in the literature [12, 24]. Key questions that any prediction modeler should ask before developing a model are:

- What is the clinical problem that the model is trying to solve?
- What is the target population for which the model is intended?
- What is the outcome that the model is trying to predict?
- What is the intended use of the model (e.g. diagnosis, prognosis, treatment selection)?
- Who will be the end-users of the model (e.g. clinicians, patients, healthcare systems)?
- What is the clinical context in which the model will be applied (e.g. primary care, hospital, low-resource settings)?
- What kind of predictors are available and routinely measured in the target population?

These questions should have a clear answers before any model is developed because they will guide the whole process.However, before embarking on the model development, the literature needs to be searched for existing models that have been developed that might answer the same questions. This step is crucial to avoid unnecessary duplication of efforts and to build on existing knowledge [25].

Develop a model or update an existing one?

This is a fundamental questions that any model developer needs to ask upfront. If there is a good quality model that has been developed for the same target population and with the same aim, it might be appropriate to first validate that model in your data. If the performance is good, updating can be considered which could entail, recalibration, revision or extension with new predictors [26]. If the performance is poor or the quality of the existing model is low, then developing a new model might be the best option. However, developing a new model is more resource intensive and requires more expertise than updating an existing one, therefore it should be considered only when there is a clear need for it. Publishing a new model without a clear justification contributes to the already large number of models in the literature and adds to the confusion among clinicians about which model to use [16]. Once the need for a model is identified, we can proceed to the next steps of model development, which include variable selection, handling missing data, model specification and fitting, and validation.

Variable selection and sample size

From the potential set of predictors, the goal of variable selection is to identify the subset of predictors that will be included in the final model. Including too many predictors can lead to **overfitting** and optimism where the model learns non-generalisable patterns (i.e. noise) and provides performance measures that are overly optimistic. Conversely, including too few can lead to **underfitting** and poor performance (i.e. missing signal). Variable selection is normally guided by clinical knowledge, previous research, and statistical criteria [27]. Common statistical methods for variable selection in logistic regression include backward elimination or regularization techniques (e.g. Lasso, ridge) although they primarily rely on statistical properties such as p-values and information criteria (e.g. Akaike information criterion or Bayesian information criterion). This focus on statistical properties may not align with the ultimate goal of improving patient outcomes because the patient-burden of measuring certain predictors is ignored. Note that the goal of variable selection is not to find the best set of predictors for the data at hand, but rather to find a set of predictors that will provide good performance in new data. Therefore, variable selection should be guided by the intended use of the model and the clinical context in which it will be applied. For instance, if the model is intended to be used in a low-resource setting, then predictors that are not routinely measured in that setting should be avoided, even if they have good statistical properties in the development data. Collaboration with clinicians and other stakeholders is crucial to ensure that the selected predictors are relevant and feasible to measure in practice [25].

Variable selection is closely related to sample size. Determining the sample size depends on several factors but the number of parameters in the model is a crucial one. The number of parameters depends on the candidate predictors; binary predictors require one parameter each, while categorical predictors with more than two categories require multiple parameters. For continuous predictors, if non-linearities are ignored, only one parameter is needed per predictor, but if non-linearities are modelled using splines or fractional polynomials, the number of parameters can increase substantially. A commonly used but not recommended rule of thumb is to include at least 10 events per predictor, but this is not a one-size-fits-all rule and more sophisticated sample size calculations should be performed [28, 29]. In general, the more parameters in the model, the larger the sample size needed to avoid overfitting and ensure reliable estimates. Flexible methods such as random forest or gradient boosting algorithms have shown to require larger sample sizes to avoid optimism [30, 31].

Handling missing data

Missing data is a common issue in clinical datasets and can lead to different types of bias if not handled properly. The most common approach is complete case analysis (CCA), which only includes patients with complete data for all predictors and the outcome [32]. This approach is pragmatic, but it leads to biased estimates unless data is **missing completely at random** (MCAR), which is often unrealistic in clinical datasets [33]. A more robust approach is imputation, which fills in missing values based on observed data and typically assumes **missing at random** (MAR). Imputation can still be biased under **missing not at random** (MNAR), where missingness depends on the unobserved value itself. The recommended approach for prediction models is multiple imputation, where missing values are imputed multiple times to create multiple complete datasets and properly account for imputation uncertainty [34]. If data are missing because a predictor is not routinely measured in the target population, excluding that predictor may be preferable to improve model applicability in practice.

Model specification, fitting and internal validation

Model specification involves choosing the functional form of the relationship between predictors and the outcome. For logistic regression, this typically involves deciding whether to include predictors as linear terms, to model non-linear relationships using splines or fractional polynomials, and whether to include interaction terms. For more complex models, such as deep learning, random forests or gradient boosting machines, model specification involves selecting the architecture and algorithm (e.g. number of layers and nodes in a neural network) and hyperparameters (e.g. learning rate, regularization strength). The process to select hyperparameters is called **hyperparameter tuning** and it is normally done by selecting the combination of hyperparameters that yields the best performance using **bootstrap** or repeated **cross-validation** among a grid of possible values for each hyperparameter [35].

Then model fitting involves estimating the parameters of the model using the available data. For logistic regression, this is done using maximum likelihood estimation, which finds the parameter values that maximize the likelihood of observing the data given the model. For more complex models, such as deep learning, random forests or gradient boosting machines, fitting involves optimizing a loss function that measures the discrepancy between observed outcomes and predicted probabilities. In the bayesian framework, fitting involves specifying prior distributions for the parameters and updating these distributions based on the observed data [36]. The choice of fitting method and model specification will affect both the performance and interpretability of the model.

Once the model is fitted, **internal validation** is the process of assessing the model's performance using the same data that was used for model development. This can be done using resampling techniques such as cross-validation or bootstrapping, which provide estimates of the model's performance that are less optimistic than the **apparent performance** obtained from the training data [12, 37]. Internal validation is crucial to detect overfitting and to obtain a more realistic estimate of how the model will perform on new data. However, it does not replace external validation, which is necessary to confirm that the model's performance generalizes to other populations and settings.

1.2.3 Dealing with clustering in model development and evaluation

Clinical prediction models are increasingly developed and validated using data from multiple centres or studies, generally known as clusters. An analysis of models in the Tufts PACE database found that 64% of models developed after 2000 used multicenter data [38]. Clustering introduces dependencies among patients within the same cluster, which can affect both model development and evaluation. Performance, particularly calibration, varies across clusters due to differences in **case mix**, measurement practices, and local prevalence [39]. Ignoring clustering during validation can lead to misleading conclusions, as overall performance summaries may mask important heterogeneity between clusters [40]. Figure 1.5 illustrates this with a simulated example: the overall calibration curve lies on the diagonal, suggesting excellent calibration, while the cluster-specific curves reveal substantial miscalibration in opposite directions. For these reasons, methods that account for clustering when developing and evaluating clinical prediction models should be developed and adopted [41, 42]. These include meta-analytic techniques to summarise performance across clusters (e.g. individual patient data (IPD) meta-analysis), and a comprehensive analysis of the between-cluster heterogeneity as outlined in the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD)-Cluster guideline.

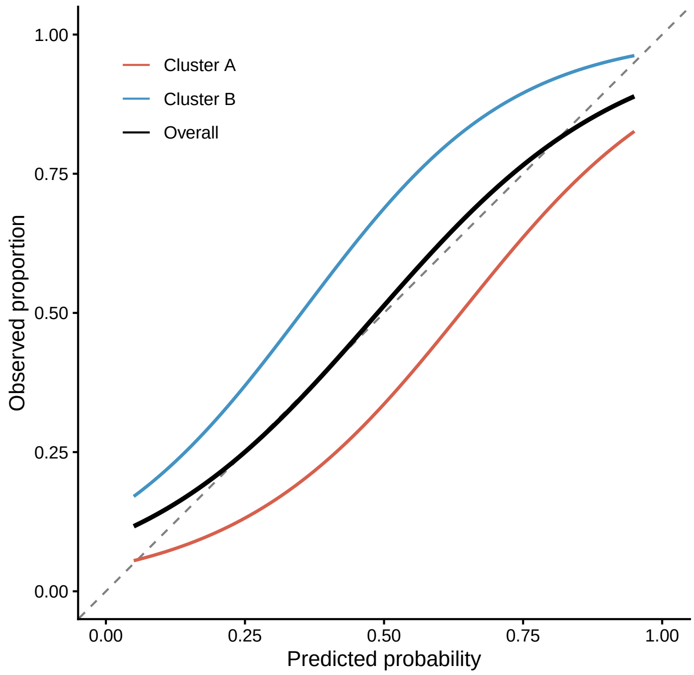


Figure 1.5: Simulated calibration curves illustrating how clustering can mask miscalibration. The overall curve (black) appears perfectly calibrated, but centre-specific curves reveal that Centre A (red) overestimates and Centre B (blue) underestimates the observed event rate.

Meta-analysis and heterogeneity

The previous section highlighted that ignoring clustering can mask important differences in model performance across centres. When a prediction model is validated in multiple clusters, whether centres in a multicentre study or studies in a systematic review, each cluster might have a different performance in terms of AUC, **calibration intercept**, **observed-to-expected ratio** or NB. These cluster-specific performances can be formally synthesised using **random effects meta-analysis** [41, 43]. In this framework, each cluster's true performance is assumed to be drawn from a distribution with overall mean μ and between-cluster variance τ^2. We assume that there is heterogeneity and that there is no single 'true' performance; instead, the true performance varies across clusters. The pooled estimate represents the expected performance in a typical cluster, while τ^2 captures how much the true performance varies across clusters beyond what is expected from sampling variability alone.

The pooled estimate is obtained as a weighted average of the cluster-specific estimates, where weights reflect both the within-cluster variance (σ_j^2) and the estimated between-cluster variance (τ^2). Several frequentist estimation methods exist for τ^2, including DerSimonian-Laird (DL), restricted maximum likelihood (REML), and Paule-Mandel (PM) [44]. In addition, the Hartung-Knapp-Sidik-Jonkman (HKSJ) adjustment replaces the standard Wald-type confidence interval with a t-distribution-based interval, providing better control of type I error rates when the number of clusters is small [45]. The HKSJ adjustment can be combined with any τ^2 estimator. As an alternative to frequentist methods, Bayesian random effects meta-analysis places a prior distribution on τ^2 and estimates the posterior distribution of the pooled effect; this approach naturally incorporates the full uncertainty about τ^2 into both the credible interval for the pooled estimate and the **prediction interval** for a new cluster, but requires specification of prior distributions and computational resources [46].

Beyond the pooled estimate, understanding and quantifying heterogeneity is central to interpreting the results of a meta-analysis. Several complementary measures are available, each capturing a different aspect of heterogeneity (Table 1.3). The between-cluster variance τ^2 is the fundamental heterogeneity parameter and is on the scale of the performance metric being pooled. The I^2 statistic expresses the proportion of total variability attributable to between-cluster heterogeneity rather than sampling error [47]. Although I^2 is widely reported, it depends on within-cluster precision and can be misleadingly high in large studies or misleadingly low in small studies, even when the absolute heterogeneity (τ^2) is the same [48]. Cochran's Q tests the null hypothesis of no heterogeneity, but has low statistical power when few clusters are available. For these reasons, the prediction interval is arguably the most clinically informative summary: it estimates the range within which the performance of the model is expected to fall in a new, similar cluster [46, 49]. The most common approach to obtaining the prediction interval was introduced by Higgins et al. [46]. It combines the pooled estimate, its standard error, and the estimated τ^2 to form the interval $\hat{\mu} \pm t_{k-2,1-\alpha/2}\sqrt{\mathsf{SE}(\hat{\mu})^2 + \tau^2}$, where $t_{k-2,1-\alpha/2}$ is the $1-\alpha/2$ percentile of the t-distribution with $k-2$ degrees of freedom and k is the number of clusters. With $\alpha = 0.05$, the resulting 95% prediction interval indicates the range within which the true performance of a hypothetical new cluster is expected to fall. However, this frequentist approximation has been shown to perform well only when heterogeneity is substantial and cluster sample sizes are reasonably similar [49]. In Bayesian meta-analysis, the prediction interval can be directly derived from the posterior distribution of the pooled effect and τ^2 and it is recommended because it ensures that the model accounts for the uncertainty of estimating $\mathsf{SE}(\hat{\mu})^2$ and τ^2 [41] .

Table 1.3: Summary of heterogeneity measures in random effects meta-analysis of prediction model performance.

Measure	Definition	Interpretation
Within-cluster variance (σ_j^2)	Sampling variability of the performance estimate within cluster j	Reflects the precision of the cluster-specific estimate; decreases with larger sample sizes
Between-cluster variance (τ^2)	Variance of the true underlying performance across clusters	Core heterogeneity parameter; represents variability in true performance beyond sampling error
I^2	Proportion of total variability due to between-cluster heterogeneity: $\tau^2/(\tau^2+\tilde{\sigma}^2)$	Ranges from 0% to 100%; depends on study precision and should not be interpreted in isolation
Cochran's Q	Weighted sum of squared deviations of cluster estimates from the pooled estimate	Tests the null hypothesis of no heterogeneity; low power with few clusters
95% Prediction interval	Interval for the expected performance in a new cluster	Incorporates both uncertainty in the pooled mean and τ^2; directly relevant for generalisability

Figure 1.6 illustrates a simulated random effects meta-analysis of the AUC across ten studies, comparing these estimation approaches. While the pooled point estimates are nearly identical across methods, the HKSJ adjustment consistently widens the confidence interval relative to the Wald-type interval for a given τ^2 estimator. The Bayesian approach yields the widest prediction interval, reflecting its fuller accounting of uncertainty in τ^2. Across all methods, the 95% prediction intervals

are substantially wider than any confidence interval, illustrating that the expected performance in a new setting is far more uncertain than the estimated mean performance.

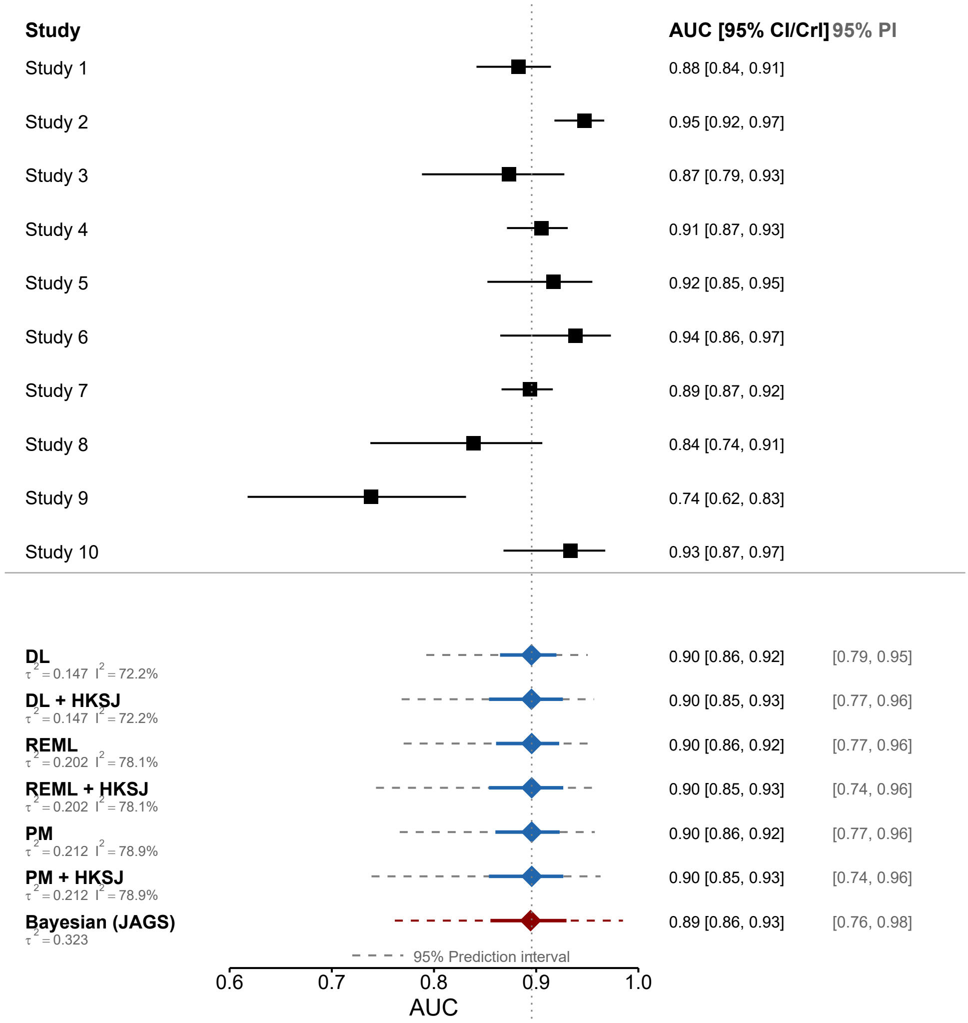


Figure 1.6: Forest plot of the AUC from ten simulated studies pooled using seven random effects meta-analysis approaches: three τ^2 estimators (DerSimonian-Laird, REML, Paule-Mandel) each with standard (Wald) and HKSJ-adjusted confidence intervals, and a Bayesian approach with a 95% credible interval. Black squares: study-specific estimates with 95% confidence intervals. Blue diamonds: frequentist pooled estimates. Red diamond: Bayesian pooled estimate using *metamisc*. Solid lines: 95% confidence or credible intervals. Dashed lines: 95% prediction intervals.

Mixed models

Another approach to deal with clustering when developing or validating a prediction model are **mixed effects models** (also known as multilevel or hierarchical models) [38, 50]. These models include fixed effects, which capture the overall relationship between predictors and outcome, and random effects, which capture cluster-specific deviations from this overall relationship. For a binary outcome, a mixed effects logistic regression with a **random intercept** per cluster j takes the form

$$\text{logit}(p_{ij}) = \beta_0 + \boldsymbol{\beta}^T \mathbf{x}_{ij} + u_j, \quad u_j \sim \mathcal{N}(0, \tau^2), \tag{1.1}$$

where p_{ij} is the probability of the event for patient i in cluster j, $\boldsymbol{\beta}^T \mathbf{x}_{ij}$ represents the fixed predictor effects, and u_j is the cluster-specific random intercept drawn from a normal distribution with variance τ^2. The random intercept allows the baseline risk to vary across clusters. The model can be extended with **random slopes** to allow the effect of one or more predictors to vary across clusters as well. If only a random intercept is included, only one extra parameter (the between-cluster variance τ^2) is added to the model, regardless of the number of clusters. In contrast, a fixed effects model (i.e. logistic regression) with a separate intercept for each cluster would require $k-1$ additional parameters. If random slopes are included, the number of parameters increases with the number of predictors that have random effects, but it still remains more parsimonious than a fixed effects model with separate parameters for each cluster.

A key consequence of including random effects is that cluster-specific estimates are shrunk towards the overall estimate due to the **shrinkage** factor. This shrinkage is determined by the between-cluster variability, within cluster variability and the number of observations per cluster. Bigger clusters that are similar to the overall estimate exhibit less shrinkage, while smaller clusters or clusters with more extreme estimates are more heavily shrunk [40]. This is desirable: it borrows information across clusters rather than treating each one in isolation, which would be imprecise for small clusters, or ignoring clustering entirely, which would mask between-cluster heterogeneity. These estimates are called **empirical Bayes estimates** because they can be interpreted as the posterior mean of the cluster-specific effect given the observed data and the estimated variance components [50].

Figure 1.7 illustrates this with a simulated example. When a standard logistic regression is fitted ignoring clustering, the predicted probabilities do not reflect the different baseline risks across clusters, resulting in poor cluster-specific calibration (Panel A). When a mixed effects model with random intercepts is used, the cluster-specific predictions are adjusted for the different baselines, bringing the calibration curves much closer to the diagonal (Panel B).

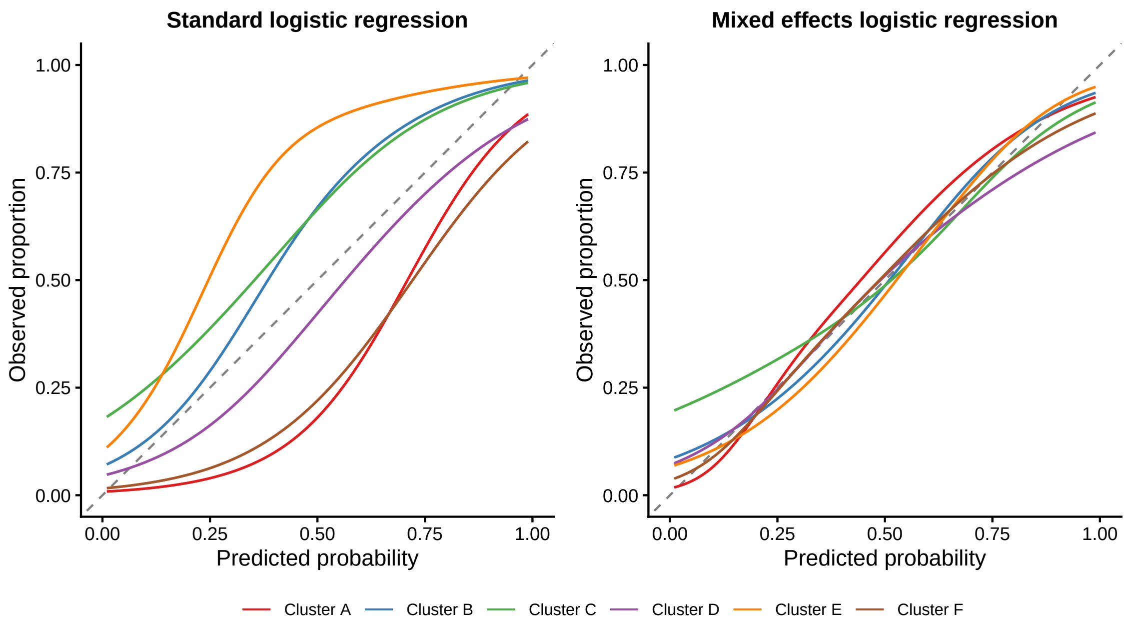


Figure 1.7: Cluster-specific calibration curves from simulated data with six clusters that differ in baseline risk. (Left) A standard logistic regression ignoring clustering produces predictions that are poorly calibrated within individual clusters. (Right) A mixed effects logistic regression with random intercepts adjusts for cluster-specific baseline risks, resulting in calibration curves that are closer to the diagonal.

Mixed models do require more complex estimation procedures and distributional assumptions (typically normality of the random effects), which should be carefully considered [50].

1.2.4 The EQUATOR network and the STRATOS initiative: enhancing the quality of prediction model research

The Enhancing the QUAlity and Transparency Of health Research (EQUATOR) network is an international network that promotes transparent and accurate reporting of health research studies [51]. The STRengthening Analytical Thinking for Observational Studies (STRATOS) initiative is a related effort that provides guidance on the design, analysis, and reporting of observational studies, including those developing and validating prediction models [52]. STRATOS aims to produce both accesible guidance for researchers with low statistical knowledge and also detailed guidance for experts in different areas related to observational studies.

Transparent and complete reporting of prediction model studies is essential for assessing the quality and applicability of published models. The TRIPOD and its update TRIPOD+AI statements provide a checklist of items that should be reported in studies developing, validating, or updating a prediction model [5, 53]. A recent extension, TRIPOD-SRMA, addresses reporting for systematic reviews and meta-analyses of prediction model studies [18]. To handle the growing use of multicenter data, TRIPOD-Cluster was published to guide the reporting of studies that develop or validate models using clustered data [42]. For the analysis of risk of bias Prediction model Risk Of Bias ASsessment Tool (PROBAST) and its update PROBAST-AI provide a structured tool to assess the risk of bias and applicability of prediction model studies [54, 55].

STRATOS has provided detailed guidance about performance measures, [19] variable selection [27], simulation studies [56], handling missing data [57] and methodological research [58] among other topics. The whole published guidance can be found in the STRATOS website (`https://www.stratos-initiative.org`) and it is an invaluable resource for researchers developing and validating prediction models.

These guidelines and recommendations play a crucial role in improving the quality and reproducibility of prediction modelling research.

1.2.5 Medical device regulation

CPMs deployed as software tools are subject to medical device regulation. Under the European Union Medical Device Regulation (MDR) (EU 2017/745), software intended for medical purposes — including diagnosis, prediction, and prognosis — is explicitly classified as a medical device, even when it operates independently of any hardware [59]. Similarly, the U.S. Food and Drug Administration regulates such tools under the Software as a Medical Device (SaMD) framework and has authorised over 1,000 AI/ML-enabled devices to date [60]. The more recent EU Artificial Intelligence Act (EU 2024/1689) introduces additional requirements for high-risk AI systems, which include most medical AI applications [61]. These regulatory frameworks reinforce the need for rigorous model development, thorough external validation, and continuous post-market monitoring. These principles align closely with the methodological standards discussed throughout this thesis and support the development of trustworthy AI in healthcare. However, the monetary cost of compliance with these regulations can be prohibitive, especially for academic researchers, that might need to turn the models into profit-generating products to compensate the costs [62]. The disalignment between open science principles and current medical device regulation is a significant challenge that needs to be addressed to ensure that valuable prediction models can be safely, widely and farily adopted in clinical practice without compromising transparency and accessibility [17].

1.3 IOTA and ovarian masses

Adnexal masses, which include ovarian tumours and other masses arising from the ovary or surrounding structures, are a common finding in gynaecological practice. In 2022 more than

300,000 new cases of ovarian cancer were diagnosed worldwide with more than 200,000 deaths [63].

Ovarian cancer is the sixth most deadly cancer in women in the USA owing this high mortality rate to its late diagnosis [64]. However, most adnexal masses are benign, and only a minority turn out to be malignant. Because the optimal management of a patient with an adnexal mass depends on the histological type of the mass, accurate preoperative characterisation is essential.

Patients with a benign mass can be managed conservatively with clinical and ultrasound follow-up, or treated with minimally invasive surgery (i.e. robotic or human laparoscopy) [65, 66]. In contrast, patients with a malignant mass benefit from management in specialised oncology centres, and the specific surgical approach may differ depending on the malignant subtype [67, 68]. Borderline tumours, early-stage primary invasive tumours, advanced primary invasive tumours, and secondary metastatic tumours each require different treatment pathways. To optimise patient triage without operating on all patients, diagnostic tools that can estimate the likelihood of malignancy from preoperative information are needed.

Transvaginal ultrasound is the primary imaging modality for characterising adnexal masses. The International Ovarian Tumour Analysis (IOTA) group is a multinational research collaboration dedicated to improving the preoperative characterisation of adnexal masses through standardised ultrasound assessment. The group began by establishing consensus definitions and measurement rules for ultrasound features [69], which standardised the sonographic description of adnexal tumours and served as the foundation for systematic data collection across centres, and have been updated 26 years later [70].

Since then, the IOTA consortium has conducted a series of large-scale prospective studies (IOTA phases 1 through 7), collecting standardised clinical and ultrasound data from more than 30,000 patients with adnexal masses at centres across multiple countries. These datasets have served as the basis for developing and validating several evidence-based diagnostic models. The following section details the specific models and tests developed to determine the preoperative management of adnexal masses.

1.4 Models and tests to diagnose ovarian masses

A multitude of mathematical models and scoring systems have been proposed over the years to support the preoperative characterisation of adnexal masses (Table 1.4). However, a comprehensive systematic review by Kaijser et al. [71] highlighted that the majority of these models suffered from poor methodological quality, small development datasets, and a lack of independent external validation. Among the tools that have demonstrated clinical utility, the Risk of Malignancy Index (RMI), published in 1990, was one of the first to be widely adopted [72]. The RMI combines a serum CA-125 level, menopausal status, and five binary ultrasound features into a single score. A threshold (commonly 200) is used to classify tumours as high or low risk for malignancy. Despite being developed on data from only 143 patients at a single hospital, the RMI has been widely adopted and remains recommended by several national clinical guidelines [68].

Biomarkers have also been proposed to support the diagnosis of ovarian cancer. The most widely used biomarker is CA-125, which is associated with ovarian cancer but it is not good at distinguishing between benign and malignant tumours alone [73]. The ROMA algorithm combines CA-125 with HE4 and menopausal status to estimate the risk of malignancy [74]. CA72.4 and CA15.3 have also been proposed to discrimnate beteween malignant subtypes [75] but cell free DNA biomarkers have not yet shown to be clinically useful for the preoperative characterisation of adnexal masses [76].

The IOTA group developed several diagnostic tools building on their standardised prospective studies. The IOTA Simple Rules use ten ultrasound features (five suggestive of malignancy and five suggestive of a benign mass) to classify tumours without requiring a calculation [77]. Years later these rules where turned into a logistic regression model with the SRRisk model [78]. The IOTA logistic regression models (IOTA logistic regression model 1 (LR1) and IOTA logistic regression model 2 (LR2)) estimate the probability of malignancy using logistic regression with ultrasound and clinical variables [79]. In 2012, the IOTA group presented the Benign Simple Descriptors, a set of four ultrasound features that are highly specific for benign tumours and can be used to rule out malignancy without the need for a risk calculation [80].

1.4.1 ADNEX model

Built on the knowledge obtained through previous studies in 2014 the IOTA group published the most comprehensive model, the ADNEX model[9]. Unlike earlier models that distinguish only between benign and malignant masses, ADNEX is a multinomial logistic regression model that estimates the probability of five tumour types: benign, borderline, stage I primary invasive, stage II–IV primary invasive, and secondary metastatic. The model uses three clinical predictors (age, serum cancer antigen 125 (CA-125), and type of centre) and six ultrasound variables (maximal diameter of the lesion, proportion of solid tissue, number of papillary projections, presence of more than 10 cyst locules, acoustic shadows, and ascites). A version without CA-125 is also available for settings where the serum marker is not routinely measured. ADNEX was developed using data from 5909 patients who underwent surgery at 24 centres across 10 countries.

Van Calster et al. [39] conducted the largest prospective external validation to date, comparing six diagnostic models in 4905 patients across 17 centres. ADNEX (with and without CA-125) and Simple Rules Risk model (SRRisk) achieved the highest discrimination (AUC 0.94), while RMI had the lowest (AUC 0.89). Calibration varied across centres for all models.

The latest recomendation is the two-step strategy, where the Benign Descriptors are applied first to identify clearly benign masses (applicable in 37% of masses where 99.3% were benign), and then ADNEX is applied to the remaining masses [81].

Since its publication, ADNEX has been recommended by the International Society of Ultrasound in Obstetrics and Gynecology (ISUOG), the European Society of Gynaecological Oncology (ESGO), and the European Society for Gynaecological Endoscopy (ESGE), among others [68]. Manufacturers of ultrasound machines have begun incorporating ADNEX directly into their devices. The model formula is published and free to use. A free website to calculate the risk was offered for 9 years but due to the MDR it is not available anymore.

Despite this widespread adoption and the robust evidence supporting its performance, ADNEX model showed important heterogeneity in calibration across centres, with some centres showing substantial underestimation and others showing overestimation of the observed event rate [39]. This highlights the need for ongoing monitoring of model performance after deployment and for methods to account for clustering when developing and evaluating clinical prediction models.

1.4.2 Machine and deep learning models

In recent years, there has been a surge of interest in applying machine learning and deep learning techniques to the problem of diagnosing adnexal masses [83, 84]. These models use highly flexible algorithms (e.g., random forests, neural networks, transformers) and may incorporate a wider range of predictors, including MRI scans or raw ultrasound images or volumes. However, while some studies report promising results, many of these models have been developed and internaly validated on small datasets from single centres and few show proper external validation [84]. Most of these models are at high risk of bias and they lack generalizability. Ledger et al. [85] compared six different machine learning algorithms for the multiclass prediction of ovarian malignancy and showed that, although most are clinically useful, the choice of the algorithm alone introduces substantial prediction uncertainty for individual patients. Recently, ADNEX-AI was published using 816 ultrasound images, as a deep learning segmentation model that characterizes the adnexal mass and estimates the ADNEX feature values directly from the ultrasound images, without the need for manual feature extraction [86]. Image analysis shows promise, but bigger multicenter image and

Table 1.4: Non exhaustive overview of diagnostic models and tests for preoperative diagnosis of adnexal masses. In bold ADNEX model predictors; US = ultrasound.

Model/Test	Year	Predictors (n)	Notes
Prediction models (probabilistic output)			
LR1 [79]	2005	Personal history of ovarian cancer, hormonal therapy, **age**, **max. diameter of the lesion**, pain, **ascites**, **papillary flow**, entirely solid tumour, max. diameter of the solid part, irregular walls, **shadows**, colour score (9)	Logistic regression
LR2 [79]	2005	**Age**, **ascites**, blood flow, max. diameter of the solid part, irregular walls, **shadows** (6)	Simplified version of LR1
ROMA [74]	2008	HE4, **CA-125**, menopausal status (3)	Serum biomarker algorithm
ADNEX [9]	2014	**Age**, **CA-125**, **centre type**, **max. diameter lesion**,**proportion of solid tissue**, **papillary projections**, **>10 locules**, **shadows**, **ascites** (9)	Multinomial logistic regression (Benign, Borderline, Stage I, Stage II-IV and Metastatic)
SRRisk [78]	2016	Simple Rules features + **type of center**	Logistic regression. Adaptation of simple rules to a probabilistic approach
Two-step strategy [81]	2023	Step 1: BDs; Step 2: **ADNEX**	BDs applicable in 37% of masses where 99.3% were benign
Tests and rules (classification)			
RMI [72]	1990	**CA-125**, menopausal status, US score (5 binary features)	Score-based; threshold commonly 200
Simple Rules [77]	2008	10 binary US features (5 Benign rules + 5 Malignant rules)	If only benign rules apply, the tumour is deemed benign and viceversa. Applicable to ∼76% of the masses.
Benign Descriptors [80]	2012	4 benign US patterns (e.g. unilocular cyst, typical dermoid, typical endometrioma)	Triage tool to identify clearly benign masses from which 99.3% were benign
O-RADS [82]	2020	US morphological features (IOTA lexicon) or ADNEX risk	Risk categories (0–5)
CA-125	—	Serum **CA-125** level	Threshold commonly 35 U/mL
Subjective Assessment	—	Expert US pattern recognition	Depends on examiner experience
Histopathology	—	Microscopic examination of tissue samples	Gold standard for diagnosis

volume studies are needed to develop and then subsequently validate those highly flexible models in external image datasets before they can be reliably used in clinical practice. Image analysis poses additional challenges related to image selection, image quality heterogeneity between US vendors and standardisation of the image format. Same patient can have several images which presents clustering issues therefore there is a urging need for large datasets to train and validate the models [84].

1.5 Open science

As open as possible as close as necessary

All studies in this thesis and the thesis itself adhere to open science principles. The open science practices followed in this thesis are in line with the FAIR principles (Findable, Accessible, Interoperable, Reusable) [87] and the TOP guidelines (Transparency and Openness Promotion) [88]. All studies are published under open access licenses and deposited in public repositories. Data and code to reproduce results and figures are made publicly available as much as possible through Open Science Framework (OSF) (`https://osf.io/pgkjv/`) or GitHub (`https://github.com/LasaiBarrenada`) repositories. The computational code utilized in the preparation of the introduction, as well as the complete LaTeX typesetting source for this manuscript, are hosted in a public GitHub repository and can be accessed at `https://github.com/LasaiBarrenada/Thesis_TEX_quarto`.

I present the number of days from submission to publication and the article processing charges for each study included in the thesis to provide transparency about the publication process and to raise awareness about the unreasonable costs of open access publishing [89]. The open science practices followed in this thesis are not only a reflection of my commitment to transparency and reproducibility but also a response to the broader movement towards open science in the research community. By making studies, data and code publicly available, I aim to facilitate the verification of results, enable other researchers to build upon this work, and contribute to a more collaborative and inclusive scientific ecosystem.

I also advocate for more equitable and sustainable models of open access publishing that do not place an undue financial burden on researchers, especially those from low- and middle-income countries or early-career researchers [90, 91]. Open science should be a right, not a privilege, and the current landscape of open access publishing needs to be reformed to ensure that it is truly inclusive and accessible and not a for-profit endeavour.

1.6 Conclusion

This introduction positioned CPMs as tools to support evidence-informed decision-making by translating heterogeneous patient information into probabilistic risk estimates. It also highlighted why rigorous evaluation is essential before such probabilities are acted upon: strong discrimination alone is insufficient when decisions depend on absolute risk, calibration can vary across centres, and the practical value of a model depends on clinical utility at relevant decision thresholds.

Against this background, ovarian mass presurgical diagnosis provides a clinically important and methodologically rich case study. The remainder of this thesis therefore combines applied evidence on the performance and utility of ADNEX with methodological work on probability estimation, calibration in clustered data, and the interpretation of uncertainty in individual risk predictions. Together, these chapters aim to strengthen the evidence base and the methods needed for safe and trustworthy use of prediction models in clinical practice.

Chapter 2

Research Objectives

The primary aim of this PhD project is to improve informed decision-making in the diagnosis of ovarian cancer by updating and better understanding the ADNEX model, ensuring that it can be used safely and effectively across diverse clinical settings. The project combines applied clinical research (Part II) with methodological research (Part III). Although the applied work centres on ovarian cancer diagnosis, the methodological insights are intended to be relevant more broadly to the development and evaluation of clinical prediction models.

The applied objectives are to update, and externally validate the ADNEX model so that its performance becomes more robust and generalisable across settings. First, we systematically review and meta-analyse ADNEX model performance across all available published evidence (Chapter 3). Second, we compare head to head the ADNEX model with the RMI model to evaluate their relative performance (Chapter 4). Third, we update the ADNEX model using insights gained during more than ten years of clinical use, together with newer data that include patients who were not operated (Chapter 5).

The methodological objectives are to improve the evaluation and development of clinical prediction models, with a particular focus on accurate predicted probabilities (i.e. calibration) and on the utility of the model when accounting for misclassification costs (i.e. net benefit). To achieve this, we investigate how different algorithms and hyperparameter settings affect predicted probabilities, including understanding overfitting and the generation of probabilities in random forests (Chapter 6) and exploring the uncertainty of individual risks (Chapter 8). We then develop calibration methods and calibration plots that incorporate clustering information, thereby enabling better evaluation of model performance in studies with clustered data (Chapter 7). Finally, we explore how clinical utility can be incorporated into model development by designing a variable selection algorithm that prioritises clinical usefulness over purely statistical fit (Chapter 9).

The expected outcome is a robust, well-calibrated, and clinically useful prediction model for ovarian cancer diagnosis with demonstrated utility, together with methodological advances that support informed decision-making across clinical prediction modelling more broadly.

Part II

Applied studies: developing and validating clinical prediction models in ovarian cancer

Part III

Methodological research: obtaining accurate predictions in clinical prediction models

Part IV

Discussion and conclusion

Chapter 10

Discussion

This thesis set out to improve evidence-informed decision-making in the diagnosis of ovarian cancer by evaluating the ADNEX model and advancing statistical methodology for clinical prediction models. The work integrates an applied arm — focused on evaluating, comparing and updating ADNEX in diverse clinical settings — with a methodological arm that advances methodology in model development, evaluation and probability estimation.

10.1 Summary of findings

Chapter 3 presented the first comprehensive systematic review and meta-analysis of ADNEX external validation studies. Across 47 studies encompassing more than 17 000 tumours, the model demonstrated robust discriminative performance, with a summary AUC of 0.93 for distinguishing benign from malignant masses [92]. However, the review also exposed important weaknesses in the validation literature: 91% of validations were at high risk of bias, primarily due to small sample sizes, inadequate handling of missing data, and incomplete performance assessment. Perhaps most critically, only 4 of 47 studies reported any form of calibration assessment, making it impossible to draw conclusions about the accuracy of ADNEX's predicted probabilities across settings.

Chapter 4 extended this work with a head-to-head meta-analysis comparing ADNEX to the RMI, the most widely used alternative in clinical practice. The results were unequivocal: ADNEX consistently outperformed RMI in terms of discrimination (summary AUC 0.92 vs 0.85 in operated patients) and clinical utility. The probability of ADNEX being clinically useful in a hypothetical new centre was 96%, compared to only 15% for RMI at the 10% risk threshold [93]. These findings provide strong evidence that clinical guidelines should recommend ADNEX over RMI for the preoperative characterisation of ovarian masses. The same conclusion was drawn from the recent Risk-prediction models in postmenopausal patients with symptoms of suspected ovarian cancer in the UK (ROCkeTS) multicenter study [94, 95].

A recurring theme throughout this thesis is that calibration — the agreement between predicted probabilities and observed event rates — remains underappreciated in the clinical prediction literature. The systematic reviews in Chapters 3 and 4 showed that most validation studies of ADNEX reported discrimination and classification but ignored calibration. That imbalance is methodologically weak and clinically risky. A model can rank patients correctly and still provide systematically wrong absolute probabilities, which can be harmful when the output is used to decide whether a woman with an adnexal mass should be referred, conservatively managed, or operated on [96]. In the context of ADNEX and prediction models as decision support tools, calibration and the probability estimate is not a secondary output: it is the output that informs clinical decisions.

The same applied story motivates the most ambitious project, updating the ADNEX model to *ADNEX2* (Chapter 5). The original ADNEX development data were necessarily limited to women who underwent surgery, whereas in clinical practice the model is intended to be used to determine who to operate. We used bayesian refitting borrowing information from the original model to update ADNEX with a larger and more representative dataset that includes non-operated patients. The goal is to produce the *ADNEX2* model that is better calibrated and more robust for use in a two-step strategy that starts with the simple benign descriptors, followed by the full model for indeterminate cases. The final aim of this project is to make the model available for the clinicians and therefore *ADNEX2* is undergoing medical device regulatory approval, and will be deployed in the near future through Gynaia Inc. (`https://www.gynaia.com/`) in a safe, easy to use and regulatory compliant app.

Chapter 6 shifted attention from the applied model in ovarian cancer to the behaviour of algorithms used for probability estimation. Using both a case study in ovarian malignancy prediction and a simulation study, the chapter showed that random forests can achieve almost perfect training AUCs, which is often a sign of overfitting, and at the same time competitive test performance. The key issue is not classical overfitting in the narrow sense, but the tendency of fully grown trees to create local probability peaks and unstable risk estimates. That distinction matters because prediction models are often tuned as if classification performance were the only objective. For risk prediction, however, the quality of the probabilities themselves is central, and the default strategy of growing trees as deep as possible can be counterproductive when the goal is calibration rather than only discrimination. We recommend not using the default hyperparameters for random forests when the goal is probability estimation, and instead tuning the tree depth and minimum node size to achieve better calibration [97].

Chapter 7 addressed a different but closely related issue: calibration should not be estimated assuming independence of observations when the data come from multiple clusters. Much of the evidence used to evaluate clinical prediction models is clustered by hospital or study, yet traditional calibration plots suppress that structure. The chapter therefore proposed clustered group calibration (CG-C), two-stage meta-analysis calibration (2MA-C), and mixed model calibration (MIX-C). The main lesson is that accounting for clustering provides more accurate calibration plots. The recommended approaches, especially 2MA-C with splines and MIX-C, make that distinction visible by combining an overall calibration curve with prediction intervals where the calibration of a new centre should lie. That is a more realistic summary than a single pooled curve because it asks the clinically relevant question of whether a model remains well calibrated when it is moved to a new setting.

Chapter 8 took a step back from the technical evaluation of model performance to examine a more fundamental question: how much can we trust a risk estimate for an individual patient? The study demonstrated that individual risk estimates are far more uncertain than commonly assumed. While estimation uncertainty (due to finite sample size) decreases with larger training samples, model uncertainty (due to arbitrary modelling choices) and applicability uncertainty (due to differences in measurement procedures and populations) remain substantial. Even with 10 000 training observations, the 95% range of predicted probabilities for the same patient averaged 0.39 on the probability scale. These findings challenge the common practice of presenting a single risk estimate to patients and clinicians without acknowledging the underlying uncertainty.

Chapter 9 is still under development, but it follows naturally from the methodological chapters. If a model is judged not only by fit but also by usefulness, then predictor selection should account for the clinical consequences of including or excluding variables. That question is especially relevant in settings such as ovarian mass diagnosis, where a model may guide referral, surgery, or follow-up decisions.

10.2 Embracing uncertainty in individualised decision-making

Clinical prediction models are developed and evaluated at the population level, but their purpose is to inform decisions for individual patients [4]. The chapters in this thesis show that moving from population performance to individual decision-making adds successive layers of uncertainty.

Chapter 3 established that ADNEX performs well on average across settings. Chapter 4 showed that it compares favourably with RMI for triage. Chapter 7 showed that pooled performance can mask meaningful centre-to-centre variation. Chapter 8 then showed that uncertainty persists even after all of those population-level summaries are taken into account.

That sequence matters because the apparent precision of a risk estimate can be misleading. A patient and clinician may see one number, but that number is the product of a model that may behave differently in another centre, another patient subgroup, or another data collection context. Chapter 7 makes this visible at the centre level; Chapter 8 makes it visible at the individual level. Chapters 3 and 4 established that ADNEX performs well on average across diverse populations. However, both chapters noted considerable heterogeneity in performance across centres and studies. Chapter 7 provided formal methods to quantify this heterogeneity, showing that calibration can vary substantially between clusters even when the overall curve appears satisfactory. This has direct implications for the concept of EIDM introduced earlier in the thesis. A predicted probability should be treated as an estimate based on the used model, not as an exact measurement. Confidence intervals will represent how uncertain the model is based on the used data, but do not guarantee that the true underliying risk of the patient lies within the interval. In practice, that means risk estimates should be communicated together with an explanation of what is known, what is uncertain, and why the estimate may change if the patient were seen in another centre or assessed under slightly different assumptions [98–101]. Chapter 8 pushed this line of reasoning further, showing that even within a single centre, the predicted probability for an individual patient is subject to model and applicability uncertainty that cannot be reduced by increasing sample size alone [98, 99]. A risk estimate of 15% should not be interpreted as a precise measurement but rather as a point within a range of plausible values, each of which might lead to a different clinical decision. These findings do not diminish the value of clinical prediction models. ADNEX remains a useful tool for ovarian mass characterisation because it shows clinical utility at population level and can support referral and management decisions. The lesson is instead that its outputs should be used with caution. Clinical prediction models should support shared decision-making rather than replace it, and that requires assesments embracing this uncertainty [4, 11].

10.3 Towards honest and safe medical devices

The systematic reviews in Chapters 3 and 4 consistently identified poor methodological quality in external validation studies. This is very concerning because this evidence is of utmost importance for safe implementation of clinical prediction models [102]. The most common issues — assessed using the PROBAST tool [103] — were inadequate sample sizes, inappropriate exclusion of patients with missing data, and incomplete reporting of model performance. Adherence to the TRIPOD reporting guidelines [5] was low, with key items such as sample size justification, missing data handling, and calibration assessment reported in fewer than 8% of studies. These findings are consistent with previous reviews of clinical prediction model validation studies, which have repeatedly highlighted similar issues [14, 15, 104–108].

These findings are even more concerning in light of the regulatory developments discussed in the introduction. As clinical prediction models increasingly fall under medical device regulation — including the EU MDR and the SaMD framework — the evidence base for their validation must improve. Regulatory frameworks require demonstration of safety and effectiveness, which in the context of prediction models translates to robust evidence of discrimination, calibration, and clinical utility across diverse populations [5, 19]. The current state of the validation literature and regulatory approval, falls short of what should be expected for tools that are intended to be used in patients. Recent studies have highlighted that many models that have been approved and are in use, have not undergone rigorous external validation, and their performance in real-world settings is often unknown or unclear [109, 110].

Improving this situation requires combined efforts from researchers and regulators. Researchers should use better study designs, with adequate sample size[111], proper handling of missing data and prospective multicenter data collection. The quality of reporting in published literature also needs improving which can be done by adhering to international guidelines like TRIPOD) and its extensions [5, 18, 42]), and a shift in culture towards recognising external validation as an ongoing process rather than a one-time exercise is needed [13]. From the regulators side, there should be stricter requirements about the model's performance before approval, ensuring that at least key performance measures are presented such as the calibration plot and a clinical utility metric (i.e. Net Benefit) [19, 112]. Both perspectives are synergistic: better research leads to better evidence for regulators, and stricter regulation incentivises better research. Ultimately, the goal is to ensure that clinical prediction models are safe, effective, and trustworthy tools for improving patient care.

10.4 Future directions for trustworthy AI in medicine

There is an exponential growth in the development of clinical prediction models, but the findings of this thesis suggest that there is still much work to be done to ensure that these models are trustworthy and useful in practice. A recent systematic review showed that the probability of a new clinical prediction model being externally validated in 5 years is of 13% [16]. Only one in eight models will be validated, which is a requirement for being trustworthy and applied in clinical practice. The findings of this thesis suggest that there is need for more and better validation studies, with a focus on calibration and clinical utility. This is a fast growing and evolving field, and there are many opportunities for future research to improve the development, evaluation, and implementation of clinical prediction models. Some of the most promising directions include:

10.4.1 Focus on clinical utility

- NB as a loss function

10.4.2 Better uncertainty estimation and interpretation

- How can we obtain CI and PI for individual predictions?

10.4.3 Evaluation when clustering is present

- Better estimation of prediction intervals - expand to survival models etc.

10.4.4 Policymakers and regulators as key stakeholders

- Ensure that the state of the art recommendations in model development and validation are implemented in regulatory frameworks, and that regulators enforce these requirements in practice.

10.4.5 Trustworthy AI for ovarian cancer diagnosis

- Calibration IPDMA - Volumes - Images - Multimodal models etc.

10.4.6 Rapid evolving field

- Image analysis - Transformer models - Foundation models etc.

10.5 Conclusion

This thesis contributes to the development and evaluation of clinical prediction models for ovarian cancer diagnosis from both applied and methodological perspectives. The systematic evaluation of ADNEX confirmed its value as a diagnostic tool and provided the evidence base for recommending it over alternatives such as the RMI. At the same time, the methodological chapters showed why this should not be the end of the story: calibration is rarely assessed, probability estimates from machine learning algorithms can be misleading if not validated properly, clustering changes how performance should be interpreted, and individual risk predictions carry more uncertainty than is often acknowledged.

These findings underscore the need of more research in the area of clinical prediction models, including ADNEX. Although they are very valuable, they need to be used responsibly. Trustworthy AI requires honest model development and validation, ensuring that the results are reliable and generalisable. Their responsible use requires rigorous validation, transparent communication of uncertainty, and ongoing monitoring — principles that are increasingly formalised in both reporting guidelines and medical device regulation, although not properly enforced by some regulators [109]. We must ensure that the rapid growth in predictive AI field is matched by a commitment to rigorous evidence generation and responsible implementation, so that these tools can truly improve patient care without unintended harms [112, 113].

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Part V

Back matter